Quiz 2

Name: _____

AI 501: Mathematics for AI | Duration: 30 min | Student ID: _____
Total: 10 marks

Question 1 (3 marks)

The following vectors represent the values of two features across four observations:

$$x = (1, 2, 3, 4), \qquad y = (2, 4, 6, 8).$$

Find the correlation between x and y.

Solution: Using the Pearson correlation coefficient formula,

$$\text{corr}(x, y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

The means are

$$\bar{x} = 2.5, \qquad \bar{y} = 5.$$

Thus

$$x - \bar{x} = (-1.5, -0.5, 0.5, 1.5)$$

and

$$y - \bar{y} = (-3, -1, 1, 3).$$

Therefore,

$$\begin{aligned}\text{corr}(x, y) &= \frac{(-1.5)(-3) + (-0.5)(-1) + (0.5)(1) + (1.5)(3)}{\sqrt{(-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2}\sqrt{(-3)^2 + (-1)^2 + (1)^2 + (3)^2}} \\ &= \frac{9 + 1 + 1 + 9}{\sqrt{2.25 + 0.25 + 0.25 + 2.25}\sqrt{9 + 1 + 1 + 9}} \\ &= \frac{20}{\sqrt{5}\sqrt{20}} = \boxed{1}.\end{aligned}$$

Question 2 (2 marks)

A company's inventory is represented by a 4-vector $\mathbf{q}$, where each component q_i represents the quantity of product i in stock. The cost (in PKR) per unit of each product is given by the vector $\mathbf{c} = \begin{bmatrix} 10 \\ 25 \\ 15 \\ 30 \end{bmatrix}$. If the inventory vector is $\mathbf{q} = \begin{bmatrix} 20 \\ 10 \\ 15 \\ 5 \end{bmatrix}$, find an expression for the total value of the entire inventory and compute the total value.

Solution: The total value of the inventory is given by the dot product

$$\boxed{V = \mathbf{c}^{\mathsf{T}}\mathbf{q}} = 825\text{PKR}.$$

Question 3 (2 marks)

Let u and v be nonzero vectors such that

$$\|u\|_2 = 3, \qquad \|v\|_2 = 4, \qquad \|u + v\|_2 = 5.$$

Find $u^T v$.

Solution: Using

$$\|u+v\|_2^2 = \|u\|_2^2 + \|v\|_2^2 + 2u^T v,$$

we get

$$25 = 9 + 16 + 2u^T v.$$

Hence

$$\boxed{u^T v = 0}.$$

Question 4 (3 marks)

Draw the set of all points in $\mathbb{R}^2$ that have unit (p)-norm, for $p = 1$, $p = 2$ and $p = \infty$.